

6. Ans: D

Taking right as positive

Using conservation of momentum:

$$mu_P + 0 = mv_P + mv_Q$$

$$u_P = v_P + v_Q \text{ ----- (1)}$$

Using conservation of KE:

$$\frac{1}{2}mu_P^2 + 0 = \frac{1}{2}mv_P^2 + \frac{1}{2}mv_Q^2$$

$$u_P^2 = v_P^2 + v_Q^2 \text{ ----- (2)}$$

$$(1)^2 - (2):$$

$$(v_P + v_Q)^2 - v_P^2 - v_Q^2 = 0$$

$$2v_P v_Q = 0$$

$$\Rightarrow v_P = 0$$

and

$$\Rightarrow v_Q = u_P - v_P = u_P$$

Hence, P becomes stationary and Q moves to the right with the same speed as P. When Q collides with R, the same occurs, Q becomes stationary while R moves to the right with the same speed.

7. Ans: D